

Causal Inference with Bayesian Methods

Aisha Sallah, Roger Saumure, Tianhai Zhou, Gina Kim

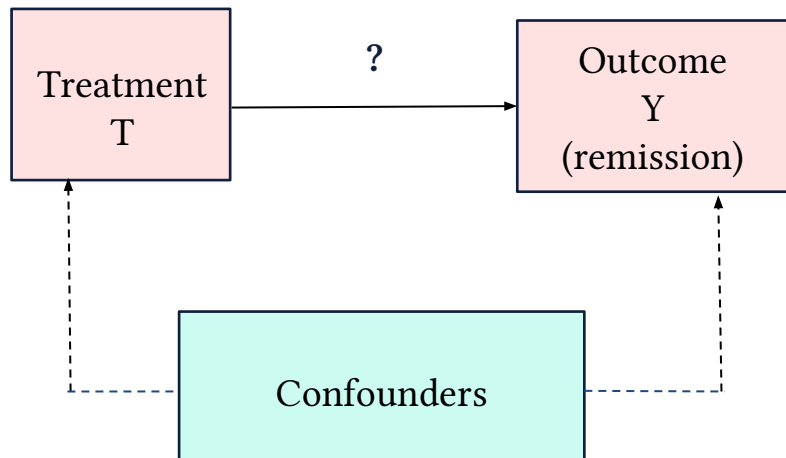
Roadmap

- Defining and Understanding Causal Inference
 - Frequentist Perspective
 - Bayesian Perspective
- Focus of today: **Synthetic Controls**
- Applied Example
 - Frequentist application
 - Bayesian Application
- Concluding Remarks

What Is Causal Inference

Correlation $\neq$ Causation

Causal inference aims to separate **true cause** from **confounding**.



Example:
Does the treatment (T) **cause** remission (Y)?

Important for **decision-making** in various fields: economics, medicine, policy business strategy, marketing...

Also for **prediction models**: which variables are causal in a process and how will intervention affect future data observations?

Frequentist Primer

Common Methods Used For Causal Inference

- Propensity Score Matching
- Differences in Differences
- Regression Discontinuity Design
- Instrumental Variables
- Synthetic Controls

Potential Outcomes Framework (Rubin-Neyman Causal Model)

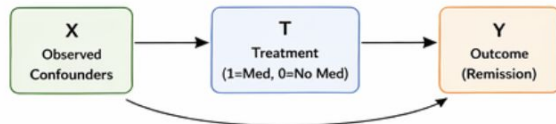
- Each unit (individual) x_i has two potential outcomes:
 - $Y_0(x_i)$ is the potential outcome had the unit not been treated:
“**control outcome**”
 - $Y_1(x_i)$ is the potential outcome had the unit been treated:
“**treated outcome**”
- Conditional average treatment effect for unit i :
$$CATE(x_i) = \mathbb{E}_{Y_1 \sim p(Y_1|x_i)} [Y_1|x_i] - \mathbb{E}_{Y_0 \sim p(Y_0|x_i)} [Y_0|x_i]$$
- Average Treatment Effect:
$$ATE = \mathbb{E}[Y_1 - Y_0] = \mathbb{E}_{x \sim p(x)} [CATE(x)]$$

Slide source: https://mlhcmiit.github.io/2023/lectures/lecture12_causal_inference2.pdf

How are we estimating the **missing** outcomes? → Causal Effects

Common Methods Used For Causal Inference

1. Propensity Score Matching (PSM)



Goal: Estimate the Average Treatment Effect (ATE) $\tau = E[Y(1) - Y(0)]$

Frequentist Model / Implementation

1 Estimate propensity scores

$$e(X_i) = P(T_i = 1 | X_i)$$

Fit logistic regression:

$$\log \frac{e(X_i)}{1 - e(X_i)} = \alpha_0 + \alpha^T X_i$$

2 Match

Match each treated unit to one or more untreated units with similar $e(X_i)$ (nearest neighbor / caliper / kernel).

3 Estimate treatment effect

$$\hat{\tau}_{ATE} = \frac{1}{N} \sum_{i=1}^N \left(Y_i^T - \sum_{j \in \mathcal{M}(i)} w_{ij} Y_j^C \right)$$

where $\mathcal{M}(i)$ is the set of matched controls for treated unit i , and w_{ij} are matching weights.

4 Inference

Standard errors via Abadie-Imbens formula or bootstrap.

Key Assumption

No unmeasured confounding: $(Y(1), Y(0)) \perp\!\!\!\perp T \mid X$
(Conditional exchangeability)

Also requires overlap: $0 < e(X) < 1$ for all X in support.

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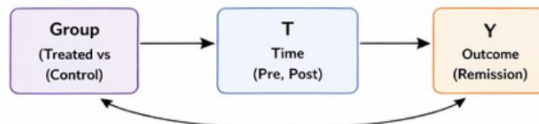
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Also requires overlap: $0 < e(X) < 1$ for all X in support.

2. Difference-in-Differences (DiD)



Goal: Estimate the Average Treatment Effect on the Treated (ATT)

$$\tau_{DiD} = E[(Y_{1,post} - Y_{1,pre}) - (Y_{0,post} - Y_{0,pre})]$$

Frequentist Model / Implementation

Regression model:

$$Y_{it} = \beta_0 + \beta_1 Treat_i + \beta_2 Post_t + \beta_3 (Treat_i \times Post_t) + \gamma^T X_{it} + \varepsilon_{it}$$

- Y_{it} : outcome for unit i at time t
- $Treat_i = 1$ if treated group, 0 if control
- $Post_t = 1$ in post-treatment period, 0 in pre-treatment
- X_{it} : optional time-varying controls

Estimate: $\hat{\tau}_{DiD} = \hat{\beta}_3$

Inference: Cluster-robust standard errors at the group level.

Key Assumption

Parallel trends: In the absence of treatment, the average change in outcomes would have been the same in treated and control groups.

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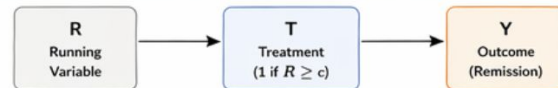
Estimate: $\hat{\tau}_{DiD} = \hat{\beta}_3$

Inference: Cluster-robust standard errors at the group level.

Key Assumption

Parallel trends: In the absence of treatment, the average change in outcomes would have been the same in treated and control groups.

3. Regression Discontinuity Design (RDD)



Goal: Estimate the treatment effect at the cutoff c

$$\tau_{RDD} = \lim_{r \downarrow c} E[Y | R = r] - \lim_{r \uparrow c} E[Y | R = r]$$

Frequentist Model / Implementation

1 Choose bandwidth h around cutoff c .

2 Fit separate local linear regressions on each side of c :

$$Y_i = \alpha_0^\pm + \alpha_1^\pm (R_i - c) + \varepsilon_i \quad \text{for } R_i \in [c - h, c + h]$$

("+" for $R_i \geq c$, "-" for $R_i < c$)

3 Estimate treatment effect: $\hat{\tau}_{RDD} = \hat{\alpha}_0^+ - \hat{\alpha}_0^-$

4 Inference: Robust (bias-corrected) standard errors (e.g., HC or clustered by running variable bins).

Key Assumptions

- No precise manipulation of R around c .
- Potential outcomes are continuous in R at c .

Common Methods Used For Causal Inference

4. Instrumental Variables (IV)



Goal: Estimate the causal effect of T on Y using instrument Z

$$\tau_{IV} = \frac{Cov(Z, Y)}{Cov(Z, T)} \quad (\text{Wald estimand in linear model})$$

Frequentist Model / Implementation (Two-Stage Least Squares)

First Stage:

$$T_i = \pi_0 + \pi_1 Z_i + \pi^T X_i + u_i$$

Obtain fitted values: $\hat{T}_i$.

Second Stage:

$$Y_i = \beta_0 + \beta_1 \hat{T}_i + \gamma^T X_i + \epsilon_i$$

Estimate: $\hat{\tau}_{IV} = \hat{\beta}_1$

Inference:

Use robust (heteroskedasticity-consistent) standard errors.

Key Assumptions

- 1. Relevance:** $Cov(Z, T) \neq 0$ (instrument affects treatment).
- 2. Exclusion Restriction:** Z affects Y only through T : $Y(1, z) = Y(1, z')$ for all z, z' .
- 3. Independence (or Conditional Independence):** $(Y(1), Y(0)) \perp\!\!\!\perp Z \mid X$ (or unconditionally if no X).

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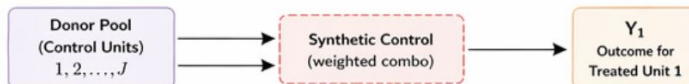
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5. Synthetic Control (SC)



Goal: Construct a weighted combination of donor units that best approximates the counterfactual outcome for the treated unit.

Frequentist Model / Implementation

Construct synthetic control:

Choose weights $w = (w_1, \dots, w_J)$ to minimize pre-treatment prediction error:

$$\min_w \sum_{t \in T_0} \left(Y_{1t} - \sum_{j=1}^J w_j Y_{jt} \right)^2$$

$$\text{subject to } w_j \geq 0, \sum_{j=1}^J w_j = 1$$

where T_0 = pre-treatment periods.

Estimate effect:

For each post-treatment period $t \in T_1$,

$$\hat{\tau}_t = Y_{1t} - \sum_{j=1}^J w_j^* Y_{jt}$$

where w^* are the optimal weights from the pre-treatment fit.

Inference:

Placebo (in-time) tests: Re-estimate using each donor as "treated" unit and compare distribution of effects. Also use permutation / randomization inference.

Key Assumption

In the absence of treatment, the treated unit would have followed the same trajectory as its synthetic control.

Today's Focus

Bayesian Primer

Why not OLS or DiD

OLS

Needs many treated units to estimate an average effect.

Cannot construct a counterfactual trajectory.

Estimates association — not causation

Diff-in-differences

Parallel trends rarely holds for one unique unit vs. many donors.

Weights all control units equally — no principled selection.

Untestable assumptions with a single unit

Synthetic control

Builds a data-driven weighted average of donor units.

Pre-treatment fit replaces parallel trends — falsifiable.

Constructs $Y(0)$ directly from the data

Synthetic Control

01

Define the donor pool

Collect untreated units that are plausibly comparable — similar in size, context, and pre-treatment trends.

02

Optimize the weights

Find weights $W = (w_1 \dots w_k)$, non-negative and summing to 1, that minimize pre-treatment fit error.

03

Read the post-treatment gap

Apply weights post-treatment. The gap between actual and synthetic is the estimated causal effect.

KEY PROPERTIES

No parallel trends required

Pre-treatment fit replaces the assumption. Poor fit is visible — and disqualifying.

Weights are explicit

Unlike DiD, SC tells you exactly which donors contribute and how much.

Single treated unit only

SC is designed for this case. It does not generalize to many treated units.

The potential outcomes framework

Causal effect = $Y(1) - Y(0)$ where only one of $Y(1)$ or $Y(0)$ is observed

$Y(1)$ – potential outcome under treatment

What we observe for the treated unit

The actual outcome for the unit that received the intervention – e.g., vaccination rate in a state that received financial support.

This is directly observable in the data.

Observed ✓

$Y(0)$ – potential outcome without treatment

The counterfactual we never see

What would have happened to the treated unit had it never received the intervention?

This is the fundamental missing data problem of causal inference – $Y(0)$ for the treated unit is unobserved.

Never observed ✗

Synthetic control solution

Estimate $Y(0)$ from donor units

SC constructs an estimate of $Y(0)$ by finding a weighted combination of untreated units (the donor pool) that closely matches the treated unit's pre-treatment outcomes.

The post-treatment gap is the estimated causal effect.

Estimated from donors

Bayesian Inference

IDENTIFICATION — synthetic control

Can we recover $Y(0)$ credibly?

Yes — if the synthetic control closely matches the treated unit before the intervention, the post-treatment gap is credibly causal.

**Bayes cannot rescue a poor pre-treatment fit.
Design quality must come first.**

ESTIMATION — Bayesian inference

How uncertain is our $Y(0)$ estimate?

Bayes gives a full posterior distribution over $Y(0)$ — capturing uncertainty from weight estimation, donor selection, and model assumptions.

Frequentist SC → *one trajectory*

Bayesian SC → *a band of plausible counterfactuals*

Bayesian Model

MODEL

Likelihood

$$Y_{it} \sim \text{Normal}(\sum w_t \cdot Y_{0t}, \sigma^2)$$

Observed outcome = weighted donor average + noise

Priors

Weights w $\sim \text{Dirichlet}(\alpha)$ — sums to 1, ≥ 0

Noise σ^2 $\sim \text{InvGamma}(a, b)$

BAYESIAN NUANCE

The Dirichlet prior on weights

α controls concentration. High $\alpha \rightarrow$ spread evenly.
Low $\alpha \rightarrow$ few donors dominate. This is your key modeling choice.

Pre-treatment fit = identification

Fit the model on pre-treatment data only. Then project weights forward to estimate $Y(0)$ post-treatment.

What changes under different α ?

Sensitivity analysis over α tests robustness. This directly answers the rubric question about assumption sensitivity.

Bayesian Synthetic Control – A More Formal Framework

Let unit 1 be treated, donor units $j = 2, \dots, J + 1$, and pre-treatment periods $t = 1, \dots, T_0$.

Frequentist SC

Choose weights on the simplex

$$\hat{\mathbf{w}} = \arg \min_{\mathbf{w} \in \Delta^J} \sum_{t \leq T_0} \left(Y_{1t} - \sum_{j=2}^{J+1} w_j Y_{jt} \right)^2$$

Counterfactual and effect

$$\hat{Y}_{1t}(0) = \sum_j \hat{w}_j Y_{jt}, \quad \hat{\tau}_t = Y_{1t} - \hat{Y}_{1t}(0)$$

Inference

Usually placebo / permutation tests;
typically yields one synthetic path.

Bayesian SC

Likelihood

$$Y_{1t} \mid \alpha, \mathbf{w}, \sigma^2 \sim \text{Normal} \left(\alpha + \sum_{j=2}^{J+1} w_j Y_{jt}, \sigma^2 \right), \quad t \leq T_0$$

Priors

$$\mathbf{w} \sim \text{Dirichlet}(\gamma), \quad \alpha \sim \text{Normal}(0, s_\alpha^2), \quad \sigma^2 \sim \text{InvGamma}(a, b)$$

Posterior

$$p(\alpha, \mathbf{w}, \sigma^2 \mid \text{data}) \propto p(Y_{pre} \mid \alpha, \mathbf{w}, \sigma^2) p(\mathbf{w}) p(\alpha) p(\sigma^2)$$

Posterior predictive counterfactual

$$p(Y_{1t}(0) \mid \text{data}) = \int p(Y_{1t}(0) \mid \theta) p(\theta \mid \text{data}) d\theta, \quad t > T_0$$

01

Weights:

optimization solution vs
posterior distribution

02

Output:

one synthetic trajectory vs
posterior band for $Y(0)$

03

Uncertainty:

placebo tests vs
credible intervals and $P(\tau_t > 0 \mid \text{data})$

What changes relative to frequentist SC?

Analysis

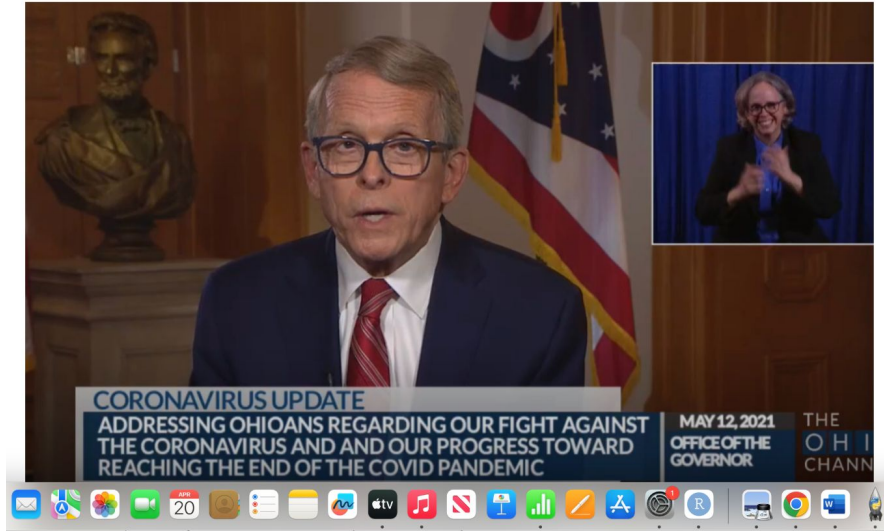


“Vax-a-Million” lottery

- Every Ohio adult who got at least one COVID-19 vaccine dose would be entered into a drawing for a \$1 million prize

Vax-A-Million seen as unorthodox, but legal way to promote vaccines in Ohio

BY: TYLER BUCHANAN - MAY 14, 2021 1:00 AM



**RQ: Did Ohio's Vax-a-Million lottery
increase COVID-19 vaccination rates
among Ohio adults?**

The Problem: can't compare pre/post

- Vaccinations were rising everywhere in Spring 2021
 - Supply expanding
 - Delta variant fears were growing
 - Employers and universities pushing vaccinations
- Even without a lottery, Ohio's rate would have kept climbing
 - A before/after comparison would attribute all that natural growth to the lottery, overstating its effect.
- Solution?
 - **What we need is a counterfactual: what would Ohio's vaccination rate have looked like if the lottery had never happened?**
 - **We can't observe this directly, however → Need to estimate a counterfactual from data**

The Synthetic Control Solution

- Find other states that didn't run lotteries but otherwise looked similar to Ohio
 - Demographics, politics, vaccination trajectory
- Use those states to construct a “synthetic” Ohio
- How synthetic Ohio is built:
 - Algorithm constructs a **weighted combination** of non-lottery donor states
 - Maybe Synthetic Ohio = 30% Indiana + 25% Pennsylvania + 20% Missouri + 15% Kentucky + 10% Michigan
 - Weights are chosen so that the weighted average tracks Ohio's pre-May-12 vaccination trajectory as closely as possible, while also matching Ohio's demographics and politics.

The Synthetic Control Solution - Continued

- Once we have Synthetic Ohio, we extend it past May 12 using the same weights applied to donor states' post-May-12 vaccination rates
- This gives us two trajectories:
 - Actual Ohio's vaccination rate after May 12 (with the lottery)
 - Synthetic Ohio's vaccination rate after May 12 (the counterfactual, no lottery)

A Frequentist Example

Data from Fuller et al. 2022

- Daily vaccination rates for all 51 jurisdictions (Ohio + donor states) over time
- State characteristics to match on
 - Demographics
 - 2020 Biden vote share
 - Income
 - Race
 - Population density... and 8 more
- **Treatment date for Ohio:** May 12, 2021
- **Donor states to create the counterfactual:** states that did *not* run lotteries

Dataset in Greater Detail – CDC County Vaccination Data

What is the unit of observation?

- **County-day panel:** one row per county (or county-equivalent) per date
- **Identifiers:** Date, FIPS, Recip_County, Recip_State, and MMWR_week
- **Coverage:** U.S. county-level COVID-19 vaccination administration and equity data
- **Source scope:** all vaccine partners, including jurisdictional clinics, retail pharmacies, long-term care facilities, dialysis centers, FEMA / HRSA partner sites, and federal entity facilities

Key variables

Vaccination outcomes

- Dose 1 counts and population percentages
- Series complete counts and population percentages
- Booster doses and vaccinated percentages

Age-specific fields

- Breakouts for 5+, 12+, 18+, and 65+ populations

Equity / context fields

- Completeness_pct
- SVI_CTGY and SVI-based vaccination measures
- Metro_status and urban-rural equity measures
- 2019 Census denominators: total, 5+, 12+, 18+, and 65+ population

01

Granularity:

daily county observations allow flexible pre/post treatment windows.

02

Aggregation:

county data can be aggregated to state-level outcomes for synthetic-control analysis.

03

Caveats:

reporting delays, revisions, and occasional unknown FIPS records mean cleaning decisions matter.

Practical implications for our study

Model

I. Define Counterfactual

$$Y_{0t}^{\text{synth}} = \sum_{j=1}^J w_j \cdot Y_{jt}$$

with

$$w_j \geq 0$$

and

$$\sum_j w_j = 1$$

II. Choose weights to match pre-period

$$\min_{\mathbf{w}} \sum_{t=1}^{T_0} \left(Y_{0t} - \sum_{j=1}^J w_j Y_{jt} \right)^2$$

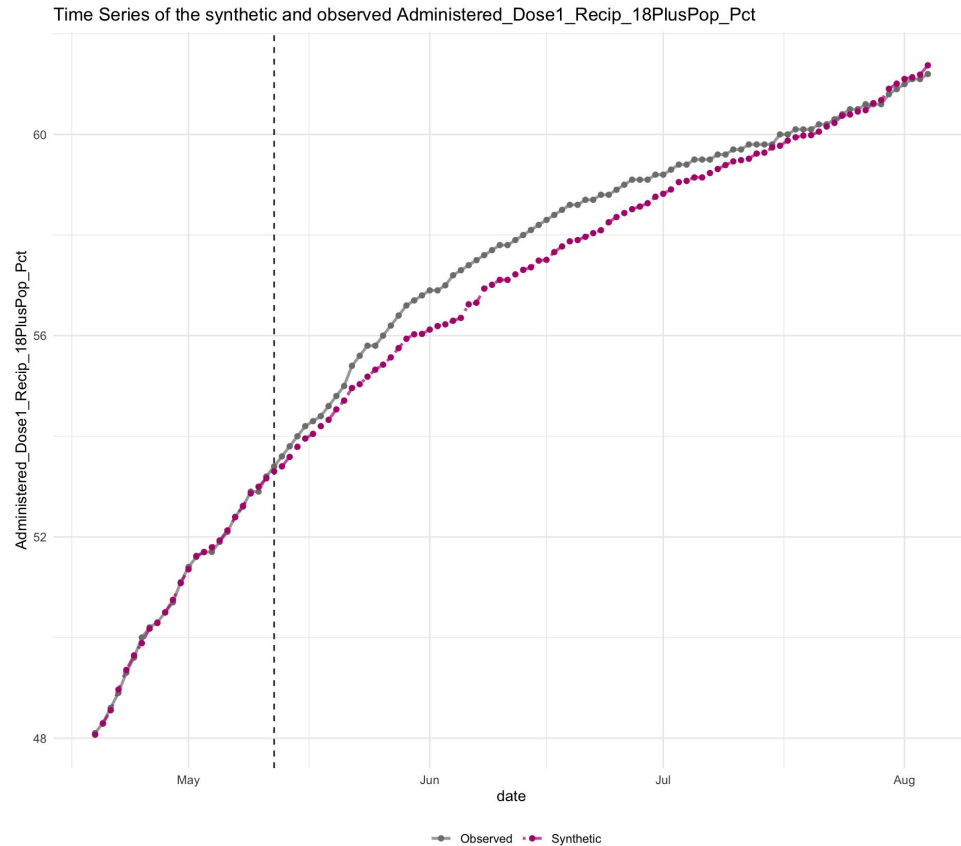
Model Continued

III. Estimate Treatment Effect

$$\hat{\alpha}_t = Y_{0t} - Y_{0t}^{\text{synth}} \quad \text{for } t > T_0$$

$$\text{TADD} = \frac{1}{T - T_0} \sum_{t=T_0+1}^T \hat{\alpha}_t$$

Results



TADD = 0.3966 → vaccination rate was 0.40 percentage points higher in Ohio

Bayesian Approach

Bayesian Synthetic Control Method

- Frequentist limitations:
 - Constraints don't fit reality (why weights need to sum to 1?)
 - No uncertainty around the weights
 - No proper way to do statistical inference (need to do placebo tests)
 - Struggles when there are many donors and few pretreatment days
- Bayesian advantages:
 - No constraints on weights
 - Priors (e.g., horseshoe) can regularize so that we can find donors that “matter”
 - Can do **statistical inference**
 - Get a posterior distribution
 - Can compute credible intervals etc.

Bayesian Synthetic Control Method

Journal of Marketing Research



Impact Factor: **5.0** / 5-Year Impact Factor: **8.1**

Journal of Marketing Research

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Research article

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Bayesian Synthetic Control Methods

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 Metrics and

Abstract

The authors propose a new Bayesian synthetic control framework to overcome limitations of extant synthetic control methods (SCMs). The proposed Bayesian synthetic control methods (BSCMs) do not impose any restrictive constraints on the parameter space a priori. Moreover, they provide statistical inference in a straightforward manner as well as a natural mechanism to deal with the “large p, small n” and sparsity problems through Markov chain Monte Carlo procedures. Using simulations, the authors find that for a variety of data-generating processes, the proposed BSCMs almost always provide better predictive accuracy and parameter precision than extant SCMs. They demonstrate an application of the proposed BSCMs to a real-world context of a tax imposed on soda sales in Washington state in 2010. As in the simulations, the proposed models outperform extant models, as measured by predictive accuracy in the posttreatment periods. The authors find that the tax led to an increase of 5.7% in retail price and a decrease of 5.5%~5.8% in sales. They also find that retailers in Washington overshifted the tax to consumers, leading to a pass-through rate of approximately 121%.

Model

I. Synthetic Ohio is a linear combination of donor states' vaccination rates:

II. Place a **horseshoe prior** on the weights

$$Y_{0t} = \beta_0 + \beta_1 Y_{1t} + \beta_2 Y_{2t} + \cdots + \beta_{31} Y_{31,t} + \varepsilon_t$$

$$\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$$

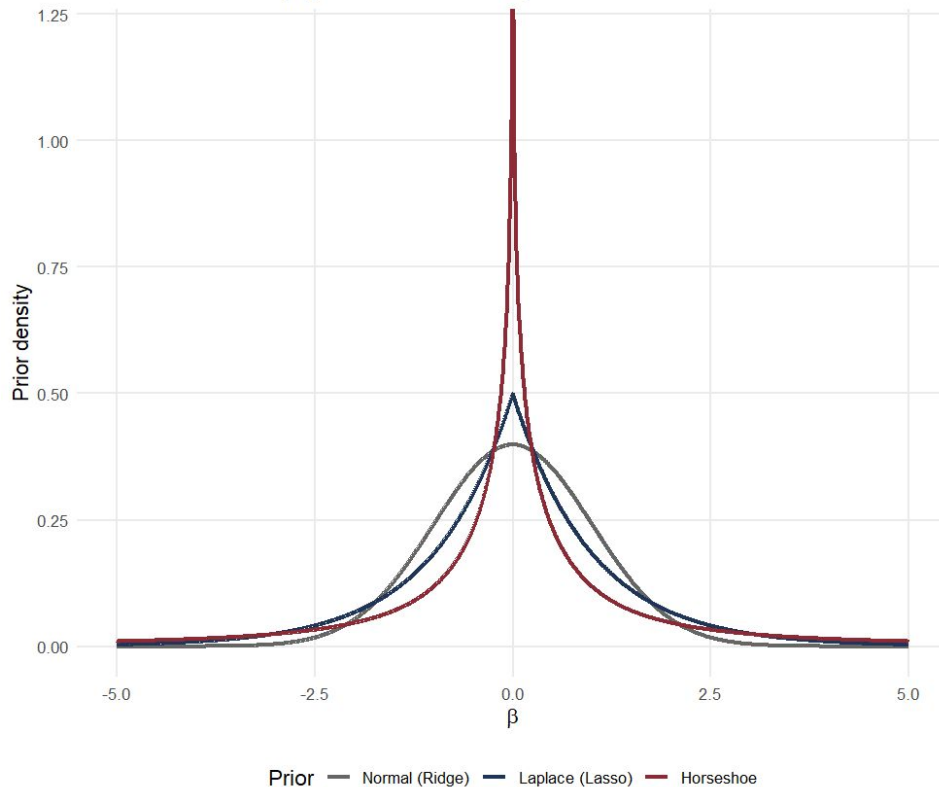
$$\beta_j \sim \mathcal{N}(0, \lambda_j^2) \quad \lambda_j \sim \text{Cauchy}^+(0, \tau)$$

$$\tau \sim \text{Cauchy}^+(0, \sigma) \quad \sigma \sim \text{Cauchy}^+(0, 10)$$

Model

Shrinkage Priors on Donor Weights

Horseshoe has a sharp spike at zero AND heavy tails



Heavy tails let important weights escape shrinkage; sharp spike squashes noise

Model - Continued

Likelihood 1 (pre-period outcome matching)

$$Y_{0t} \sim \mathcal{N}\left(\beta_0 + \sum_{j=1}^{31} \beta_j Y_{jt}, \sigma^2\right) \quad \text{for } t = 1, \dots, T_0$$

Likelihood 2 (covariate matching):

$$\mathbf{Z}_0 \sim \mathcal{N}\left(\mathbf{Z}_{\text{donors}} \boldsymbol{\beta}, \frac{1}{w_{\text{cov}}}\right)$$

Model - Continued

For each MCMC draw of β , we compute a full post-period Synthetic Ohio, then average the daily gap:

$$\alpha_t^{(s)} = Y_{0t} - \left(\beta_0^{(s)} + \sum_{j=1}^{31} \beta_j^{(s)} Y_{jt} \right) \quad \text{for } t > T_0$$

$$\text{TADD}^{(s)} = \frac{1}{T - T_0} \sum_{t=T_0+1}^T \alpha_t^{(s)}$$

Stan Code

```
stan_code_full_cov = "
data {
  int<lower=1> N_train;
  int<lower=1> N_test;
  int<lower=1> p;
  int<lower=1> K;

  vector[N_train] y_train;
  matrix[N_train, p] X_train;
  matrix[N_test, p] X_test;

  vector[K] z_treated;
  matrix[K, p] Z_donors;

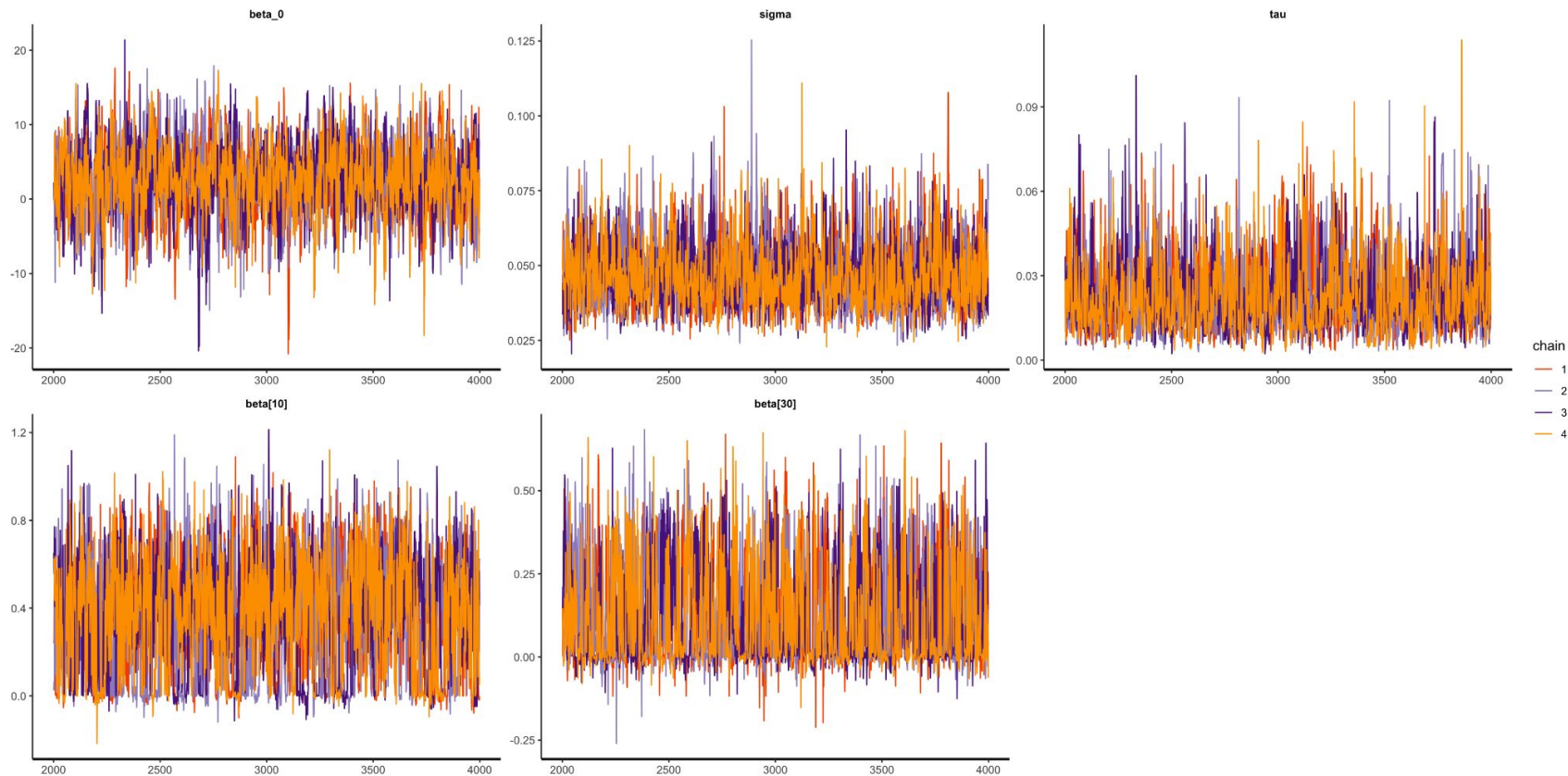
  real<lower=0> covariate_weight;
}
parameters {
  real beta_0;
  real<lower=0> sigma2;
  vector[p] beta_raw;
  vector<lower=0, upper=pi()/2>[p] lambda_unif;
  real<lower=0> tau;
}
transformed parameters {
  vector[p] beta;
  real<lower=0> sigma;
  vector<lower=0>[p] lambda;
  vector[N_train] X_beta;

  lambda = tau * tan(lambda_unif); // horseshoe local shrinkage
  for (j in 1:p) {
    beta[j] = lambda[j] * beta_raw[j];
  }
  sigma = sqrt(sigma2);
  X_beta = beta_0 + X_train * beta;
}
model {
  // Horseshoe prior
  beta_raw ~ normal(0, 1);
  tau ~ cauchy(0, sigma);
  sigma ~ cauchy(0, 10);
  beta_0 ~ cauchy(0, 10);

  // Likelihood: outcomes
  y_train ~ normal(X_beta, sigma);

  // Likelihood: covariates
  z_treated ~ normal(Z_donors * beta, 1.0 / sqrt(covariate_weight));
}
generated quantities {
  vector[N_test] y_test_pred;
  for (i in 1:N_test) {
    y_test_pred[i] = beta_0 + X_test[i,] * beta;
  }
}
"
```

MCMC Diagnostics



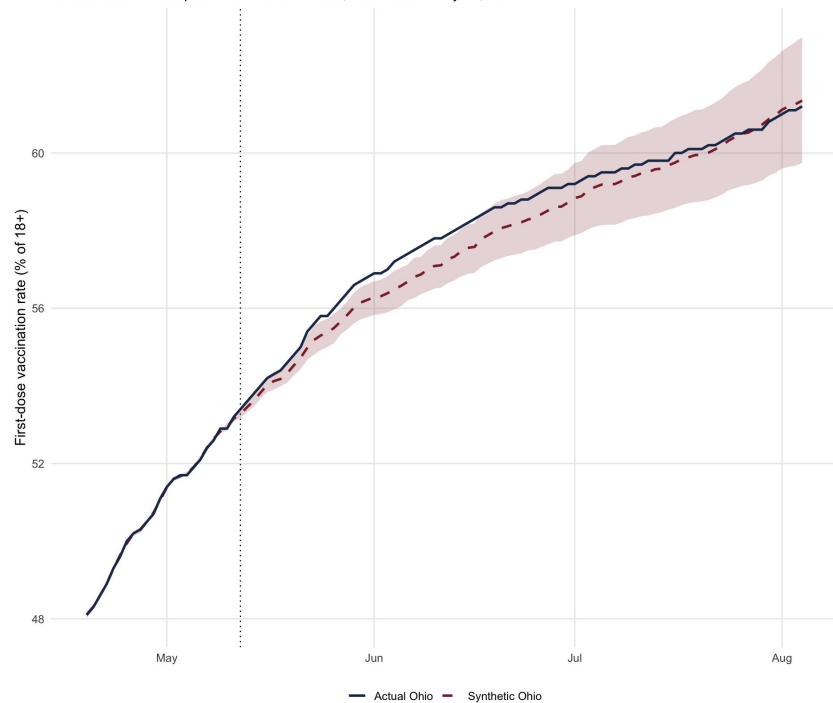
MCMC Diagnostics - Continued

Diagnostic	Value
R-hat (max)	1.005
ESS (minimum)	603
ESS (median)	1,946
Divergent transitions	1.0%

Model Results

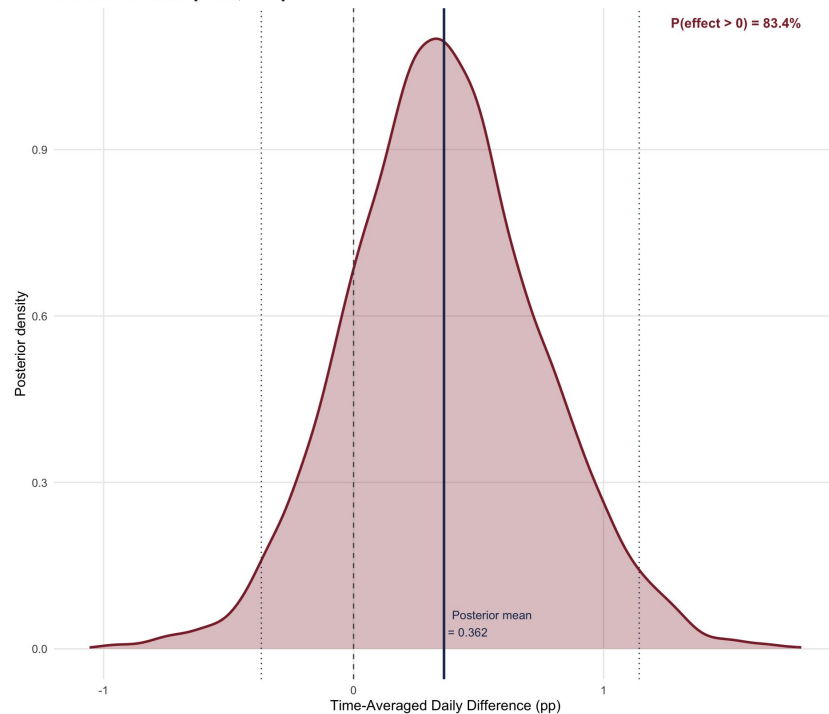
Actual vs. Synthetic Ohio

Shaded band = 95% posterior credible interval; dotted line = May 12, 2021



Posterior Distribution of Ohio Vax-a-Million Lottery Effect

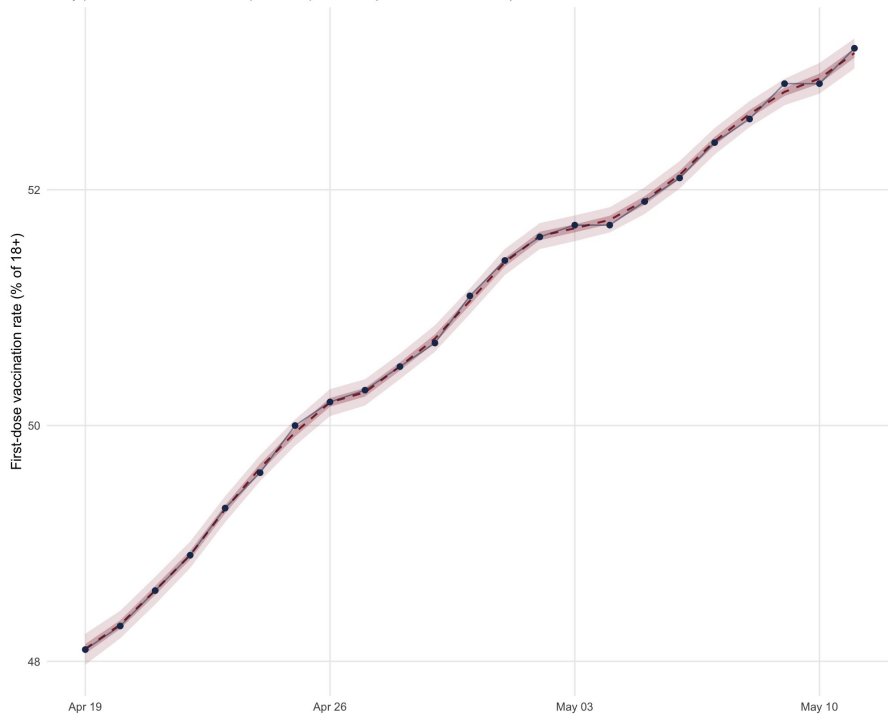
95% credible interval: [-0.369, 1.142]



Posterior Predictive Check

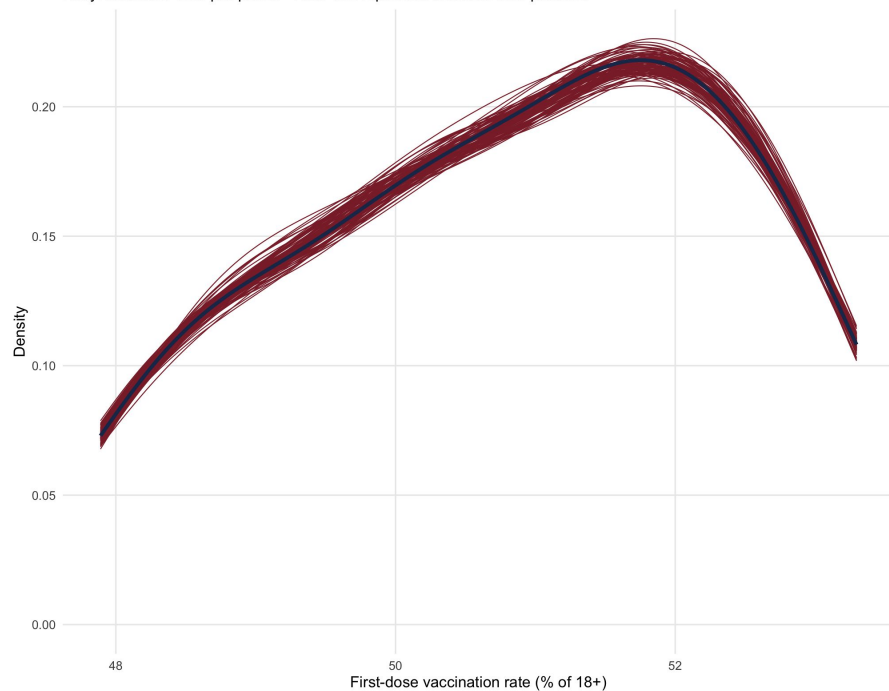
Posterior Predictive Check: Pre-Period Fit

Navy points: Actual Ohio • Red: posterior predictive (50% and 95% bands)



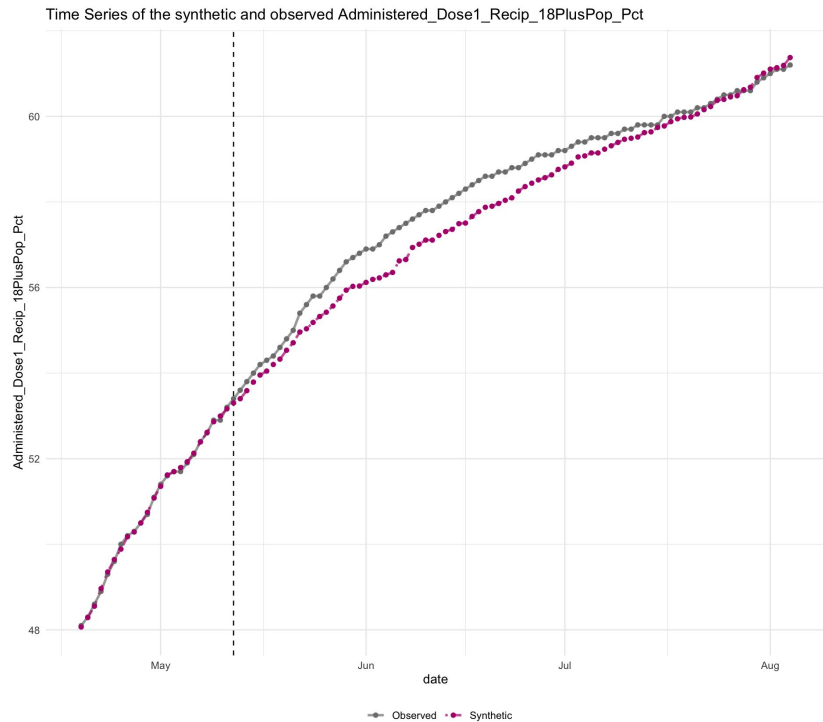
Posterior Predictive Check

Navy: observed Ohio pre-period • Red: 100 replicated datasets from posterior

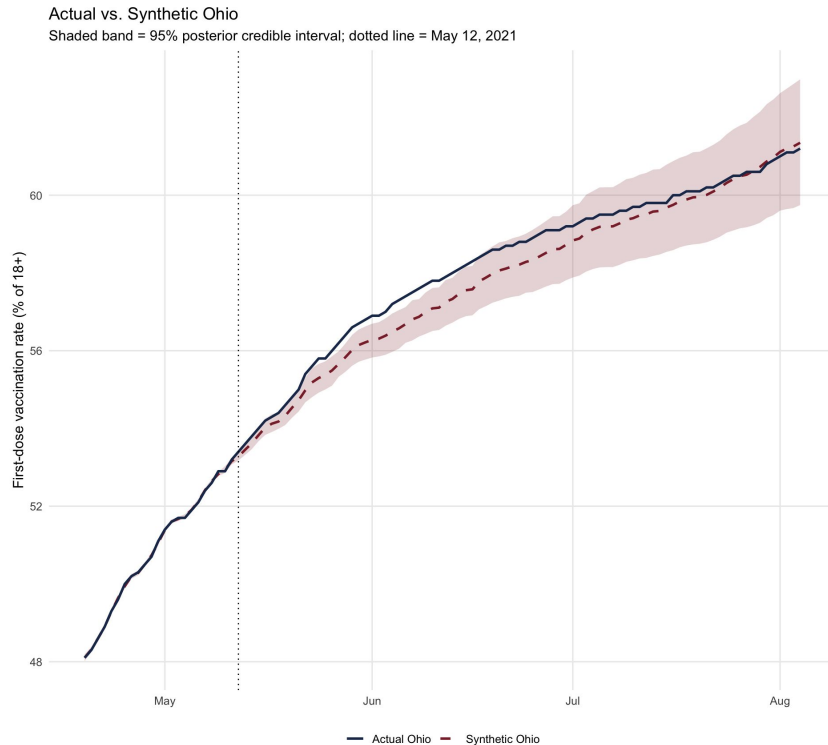


A model that fits well produces replicated datasets that look like the observed data.

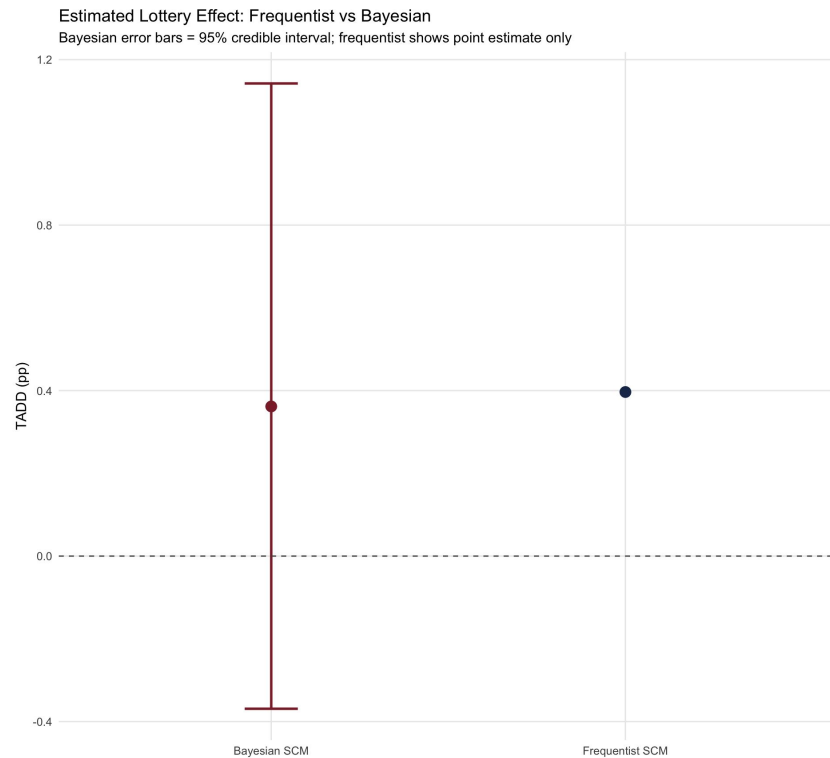
Frequentist vs. Bayesian TADD



State: OH



Frequentist vs. Bayesian TADD



Conclusion



Frequentist vs. Bayesian SCM

Frequentist SCM

Weights

Constrained: sum to 1, ≥ 0

Uncertainty

No formal uncertainty on weights

Inference

Placebo tests only — no credible intervals

Many donors

Struggles when donors > pre-treatment periods

Output

One synthetic trajectory

Bayesian SCM

Weights

Unconstrained — prior handles regularization

Uncertainty

Full posterior distribution over all weights

Inference

Credible intervals and $P(\text{effect} > 0 \mid \text{data})$

Many donors

Horseshoe prior selects relevant donors automatically

Output

A band of plausible counterfactuals

What Bayesian inference adds

- **Uncertainty is a first-class output**

Frequentist causal methods produce point estimates. Bayes produces a full posterior — you can ask any question about the distribution of the effect, not just whether it passes a threshold.

- **Priors encode domain knowledge**

Before seeing data, we often know something about plausible effect sizes or which donors are comparable. A prior lets us express that formally, rather than ignoring it or hard-coding it as a constraint.

- **Credible intervals have a direct interpretation**

A 95% credible interval means there is a 95% probability the effect lies in that range — given the data. A frequentist confidence interval does not say that, even though people often want it to.

When to use Bayesian Causal Inference

Use it when...

- You have a single treated unit with many potential donors
- You want formal uncertainty — credible intervals and posterior probabilities
- You have prior knowledge about effect sizes or donor relevance
- The pre-treatment period is short relative to the number of donors
- You need to communicate probability statements to a non-technical audience

Be careful when...

- Pre-treatment fit is poor — Bayes cannot fix bad identification
- The prior is chosen to confirm a desired result
- Computational cost matters — MCMC is slower than constrained optimization
- You need a simple, auditable method for a policy audience